Chapter 2

Mathematical modelling of bacterial resistance to multiple antibiotics and immune system response

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2.1 Overview

2.1.1 Bacterial Infections and Antibiotic Resistance

Bacterial infections remain a leading cause of disease and death globally, affecting millions of people each year across all age groups and healthcare settings. The introduction of penicillin in the 1940s marked the beginning of the antibiotic era, offering effective treatment for infections that had previously carried high mortality. Since then, numerous classes of antibiotics have been developed, each targeting different aspects of bacterial physiology such as cell wall synthesis, protein production, or DNA replication.

However, bacteria are capable of rapid evolutionary adaptation. Resistance to antimicrobial agents is now recognized as both a logical and inevitable consequence of antibiotic use. Each new class of antibiotic introduced into clinical practice has been followed within years, and sometimes months, by the emergence of resistant strains. Resistance mechanisms include enzymatic inactivation of the drug, modification of the drug target, active efflux of the drug from the bacterial cell, and reduced membrane permeability. These mechanisms can arise through spontaneous chromosomal mutations or through the acquisition of resistance genes from other bacteria via horizontal gene transfer.

2.1.2 Role of the Immune System in Bacterial Infections

The immune system is the body's primary defense against infection. It comprises two broad arms: the innate immune system, which provides a rapid and non-specific first response to pathogens, and the adaptive immune system, which mounts a targeted and specific response based on the identity of the

infecting organism. Both arms play important roles in controlling bacterial infections.

When bacteria enter the body, innate immune cells such as neutrophils and macrophages are rapidly recruited to the site of infection. These cells engulf and destroy bacteria through phagocytosis and the release of antimicrobial substances. If the innate response is insufficient to clear the infection, the adaptive immune system is activated. Specific immune cells, including T lymphocytes and B lymphocytes, are produced in response to bacterial antigens and work to eliminate the pathogen more precisely.

2.1.3 Multiple Antibiotic Therapy

The use of two or more antibiotics simultaneously, known as combination or multiple antibiotic therapy, has become standard practice for several types of bacterial infection. The rationale for this approach is threefold. First, different antibiotics act on different bacterial targets, so their effects can be additive or synergistic, producing greater bacterial killing than either drug alone. Second, combination therapy reduces the probability of resistance emerging, since a single mutation is unlikely to confer resistance to two drugs with different mechanisms simultaneously. Third, for mixed infections involving multiple bacterial species or strains with different susceptibility profiles, combinations ensure that all components of the infection are covered.

2.1.4 Mathematical Modelling in Infectious Diseases

Mathematical modelling has a long history in the study of infectious diseases. The foundational SIR model, which divides a population into susceptible, infected, and recovered compartments, was introduced in the early twentieth century and remains widely used today for studying epidemic dynamics. More sophisticated extensions of this framework have been developed to account for latent infection, vaccination, age structure, spatial heterogeneity, and many other biological complexities.

At the within-host level, models have been used to describe how bacterial populations grow and evolve under antibiotic pressure, how immune cells

respond to infection, and how drug pharmacokinetics influence bacterial killing.

2.1.5 Within-Host Models vs Epidemiological Models

Mathematical models of infectious disease can be broadly divided into two categories based on their scale of analysis. Epidemiological models, also called between-host models, describe the spread of infection across a population of individuals. These models are used to predict outbreak trajectories, evaluate vaccination programs, and assess the impact of public health interventions. The SIR model and its variants fall into this category.

Within-host models, by contrast, describe the dynamics of infection inside a single individual. These models track the populations of pathogens, immune cells, and drugs over time within one host and are used to understand the mechanisms of infection progression, immune evasion, and treatment response. Within-host models are particularly relevant for studying antibiotic resistance because resistance develops at the level of individual bacterial cells within a patient, driven by mutation and selection pressure from drugs and immunity.

2.2 Mathematical Model

2.2.1 Model Description and Assumptions

Model Description

This study examines the within-host dynamics of a bacterial infection in a single patient receiving multiple antibiotic treatment. Unlike epidemiological models such as SIR which track disease spread across a population, this model focuses on what happens inside one individual's body — specifically how sensitive bacteria, resistant bacteria, immune cells, and antibiotic concentrations interact and change over time.

The model is built as a system of ordinary differential equations (ODEs) with five state variables: the concentration of antibiotic-sensitive bacteria s(t), antibiotic-resistant bacteria r(t), host immune cells b(t), and the concentrations of two antibiotics — isoniazid INH denoted a1(t) and pyrazinamide PZA denoted a2(t). All five variables are normalized, meaning their values fall between 0 and 1, where 1 represents the maximum possible concentration or population size.

The central biological question the model addresses is: given a patient with a bacterial infection being treated with two antibiotics simultaneously, will the infection be cleared, will resistant bacteria take over, or will both types of bacteria persist alongside the immune response?

The answer depends critically on two threshold parameters A and B derived from the model parameters. Parameter A represents the net reproductive ability of sensitive bacteria after accounting for antibiotic killing and immune destruction. Parameter B represents the net reproductive ability of resistant bacteria after immune destruction. When A is less than B, only resistant bacteria survive in the long run. When A is greater than B, both sensitive and resistant bacteria coexist with immune cells at a stable equilibrium.

2.2.2 State Variables and Parameters

State Variables

The model consists of five state variables, all of which are normalized to dimensionless values in the range [0, 1]. Normalization is achieved by dividing each variable by its respective maximum capacity, which simplifies the mathematical analysis and ensures all variables remain biologically meaningful.

Variable	Meaning	Normalization	Range
s(t)	Population of antibiotic-sensitive bacteria at time t	S(t) divided by carrying capacity T	[0, 1]
r(t)	Population of antibiotic-resistant bacteria at time t	R(t) divided by carrying capacity T	[0, 1]

Variable	Meaning	Normalization	Range
b(t)	Population of host immune cells at time t	B(t) divided by ωT	[0, 1]
a1(t)	Concentration of INH (Isoniazid) at time t	A1(t) divided by δ1/μ1	[0, 1]
a2(t)	Concentration of PZA (Pyrazinamide) at time t	A2(t) divided by δ2/μ2	[0, 1]

Where T is the carrying capacity of bacteria, ω is the ratio of immune cell capacity to bacterial capacity, δi is the supply rate of the i-th antibiotic, and μi is the clearance rate of the i-th antibiotic.

Parameters

The model contains the following parameters. Each parameter is strictly positive as required for biological validity.

Parameter	Symbol	Meaning	Unit	Value Used
Sensitive bacteria growth rate	βS	Maximum per capita growth rate of sensitive bacteria	day^{-1}	0.8
Resistant bacteria growth rate	βR	Maximum per capita growth rate of resistant bacteria	day^{-1}	0.4
Fitness cost of resistance	c	Fractional reduction in growth rate due to carrying resistance genes	dimensionless	0.5
Immune killing rate	η	Rate at which immune cells kill bacteria per unit concentration	day^{-1}	0.3

Parameter	Symbol	Meaning	Unit	Value Used
Immune cell growth rate	k	Per capita proliferation rate of immune cells in response to infection	day^{-1}	0.6
Mutation rate — INH	α1	Rate at which sensitive bacteria mutate to resistant due to INH exposure	day^{-1}	0.02
Mutation rate — PZA	α2	Rate at which sensitive bacteria mutate to resistant due to PZA exposure	day^{-1}	0.06
Antibiotic killing rate — INH	d1	Rate at which INH kills sensitive bacteria	day^{-1}	0.15
Antibiotic killing rate — PZA	d2	Rate at which PZA kills sensitive bacteria	day^{-1}	0.35
INH clearance rate	μ1	Rate at which INH is cleared from the body	day^{-1}	0.06
PZA clearance rate	μ2	Rate at which PZA is cleared from the body	day^{-1}	0.03

Derived Threshold Parameters

These are not independent parameters but are calculated from the parameters above. They determine which equilibrium the system converges to.

Symbol	Formula	Biological Meaning
A	$(\beta S - \Sigma(\alpha i + di)) / (\beta S + \eta)$	Net survival fraction of sensitive bacteria after antibiotic and immune pressure

Symbol	Formula	Biological Meaning
B	$\beta R / (\beta R + \eta)$	Net survival fraction of resistant bacteria after immune pressure
C	$\Sigma\alpha i / (\beta R + \eta)$	Contribution of mutation from sensitive to resistant bacteria

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Relationship Between Parameters

$\beta R = (1 - c) \times \beta S$

meaning:

$c = 0 \rightarrow \beta R = \beta S$ (no fitness cost, resistance is free)

$c = 0.5 \rightarrow \beta R = 0.4$ (resistance costs 50% of growth rate)

$c = 1 \rightarrow \beta R = 0$ (resistance is lethal, bacteria cannot grow)

In this model $c = 0.5$ so $\beta R = (1 - 0.5) \times 0.8 = 0.4$

Notes on Units

All state variables are dimensionless after normalization. All rate parameters carry units of day^{-1} meaning they represent fractional changes per day. This is consistent with the clinical treatment timescale of the model which runs over 90 days representing a standard antibiotic treatment course.

2.2.3 The ODE System

The model is governed by a system of five ordinary differential equations describing the time evolution of sensitive bacteria, resistant bacteria, immune cells, and two antibiotic concentrations.

Equation 1 — Sensitive Bacteria

$$\frac{ds}{dt} = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot [(\alpha 1+d1) \cdot a1 + (\alpha 2+d2) \cdot a2]$$

Term	Expression	Meaning
Logistic growth	$\beta S \cdot s \cdot (1-s-r)$	Sensitive bacteria grow at rate βS but slow down as total bacteria s+r approach carrying capacity
Immune killing	$\eta \cdot s \cdot b$	Immune cells destroy sensitive bacteria at rate η proportional to both populations
Antibiotic effect	$s \cdot \Sigma(\alpha i+di) \cdot ai$	Antibiotics kill sensitive bacteria at rate di and convert them to resistant through mutation at rate αi

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Equation 2 — Resistant Bacteria

$$\frac{dr}{dt} = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot [\alpha 1 \cdot a1 + \alpha 2 \cdot a2]$$

Term	Expression	Meaning
Logistic growth	$\beta R \cdot r \cdot (1-s-r)$	Resistant bacteria grow at rate βR which is lower than βS due to fitness cost c
Immune killing	$\eta \cdot r \cdot b$	Immune cells kill resistant bacteria at the same rate η as sensitive bacteria
Mutation gain	$s \cdot \Sigma\alpha i \cdot ai$	Sensitive bacteria mutate into resistant ones upon antibiotic exposure — this is a gain for r and a loss for s

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Equation 3 — Immune Cells

$$\frac{db}{dt} = k \cdot b \cdot (1 - b/(s+r))$$

Term	Expression	Meaning
Logistic growth	$k \cdot b \cdot (1 - b/(s+r))$	Immune cells proliferate at rate k with carrying capacity equal to total bacterial load s+r. When bacteria decrease the immune response also contracts. When s+r = 0 immune cells decay at rate k

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Equation 4 and 5 — Antibiotic Concentrations

$$\frac{da1}{dt} = \mu 1 \cdot (1 - a1)$$

$$\frac{da2}{dt} = \mu 2 \cdot (1 - a2)$$

Term	Expression	Meaning
INH dynamics	$\mu 1 \cdot (1 - a1)$	INH is supplied continuously and cleared at rate μ1 driving concentration toward normalized steady state of 1
PZA dynamics	$\mu 2 \cdot (1 - a2)$	Same structure as INH with its own clearance rate μ2

Complete System

$$\frac{ds}{dt} = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot [(\alpha 1+d1) \cdot a1 + (\alpha 2+d2) \cdot a2]$$

$$\frac{dr}{dt} = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot [\alpha 1 \cdot a1 + \alpha 2 \cdot a2]$$

$$\frac{db}{dt} = k \cdot b \cdot (1 - b/(s+r))$$

$$\frac{da1}{dt} = \mu 1 \cdot (1 - a1)$$

$$\frac{da2}{dt} = \mu 2 \cdot (1 - a2)$$

Biological Constraints

The following constraints must hold at all times for the system to remain biologically valid:

Constraint	Meaning
$s(t) \geq 0$	Sensitive bacteria cannot be negative
$r(t) \geq 0$	Resistant bacteria cannot be negative
$0 \leq b(t) \leq s+r$	Immune cells are bounded by bacterial load
$0 \leq ai(t) \leq 1$	Antibiotic concentrations are normalized
$s(t) + r(t) \leq 1$	Total bacteria cannot exceed carrying capacity

2.2.4 Normalization of Variables

Why Normalize

The original model variables operate on very different scales. Bacterial populations S(t) and R(t) are measured in number of cells and can reach values as large as 10^9 bacteria. Immune cell populations B(t) operate on a different scale entirely, and antibiotic concentrations Ai(t) are measured in mg/kg. Working with variables of such different magnitudes makes both mathematical analysis and numerical simulation unnecessarily complicated.

Normalization rescales all variables to dimensionless quantities between 0 and 1 by dividing each variable by its natural maximum value. This simplifies the algebra, reduces the number of independent parameters, and makes the system easier to analyze and interpret.

Original Unnormalized System

The original system before normalization is:

$$\frac{dS}{dt} = \beta S \cdot S \cdot (1 - (S+R)/T) - \lambda \cdot S \cdot B - S \cdot \Sigma\, \bar{\alpha}i \cdot Ai - S \cdot \Sigma\, \bar{d}i \cdot Ai$$

$$\frac{dR}{dt} = \beta R \cdot R \cdot (1 - (S+R)/T) - \lambda \cdot R \cdot B + S \cdot \Sigma\, \bar{\alpha}i \cdot Ai$$

$$\frac{dB}{dt} = k \cdot B \cdot (1 - B/(\omega(S+R)))$$

$$\frac{dAi}{dt} = \delta i - \mu i \cdot Ai \text{ for } i = 1, 2$$

Where:

Symbol	**Meaning**
S(t)	Sensitive bacteria population (cells)
R(t)	Resistant bacteria population (cells)
B(t)	Immune cell population (cells)
Ai(t)	Concentration of i-th antibiotic (mg/kg)
T	Carrying capacity of bacteria (maximum population)
λ	Per capita immune killing rate
ω	Ratio of immune cell capacity to bacterial capacity
δi	Constant supply rate of i-th antibiotic
μi	Clearance rate of i-th antibiotic
ᾱi	Mutation rate due to i-th antibiotic
d̄i	Death rate of sensitive bacteria due to i-th antibiotic

Substitution Step

We introduce the following normalized variables:

$s(t) = \frac{S(t)}{T}$, $r(t) = \frac{R(t)}{T}$, $b(t) = \frac{B(t)}{\omega T}$, $ai(t) = \frac{\mathrm{Ai(t)}}{\delta \mathrm{i}/\mu \mathrm{i}}$

With the new variables, the normalized system is given as:

$$\frac{ds}{dt} = \beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot [(\alpha 1+d1) \cdot a1 + (\alpha 2+d2) \cdot a2]$$

$$\frac{dr}{dt} = \beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot [\alpha 1 \cdot a1 + \alpha 2 \cdot a2]$$

$$\frac{db}{dt} = k \cdot b \cdot (1 - b/(s+r))$$

$$\frac{da1}{dt} = \mu 1 \cdot (1 - a1)$$

$$\frac{da2}{dt} = \mu 2 \cdot (1 - a2)$$

The normalization step reduces the number of independent parameters by combining them:

New Parameter	Formula	Meaning
η	λ · ω · T	Combined immune killing rate in normalized system
αi	ᾱi · (δi/μi)	Normalized mutation rate for i-th antibiotic
di	d̄i · (δi/μi)	Normalized killing rate for i-th antibiotic
βR	(1−c) · βS	Resistant growth rate incorporating fitness cost

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2.2.5 Threshold Parameters A, B and C

Definition

Three threshold parameters are derived from the model parameters. These parameters determine which equilibrium the system converges to and therefore determine the long-term outcome of the infection.

Parameter	Formula	Meaning
A	$(\beta S - \Sigma(\alpha i + di)) / (\beta S + \eta)$	Net Survival Ability of Sensitive Bacteria
B	$\beta R / (\beta R + \eta)$	**Net Survival Ability of Resistant Bacteria**
C	$\Sigma\alpha i / (\beta R + \eta)$	**Mutation Contribution**

Written explicitly for two antibiotics:

$$A = \frac{\beta S - (\alpha 1 + d1) - (\alpha 2 + d2)}{\beta S + \eta}$$

$$B = \frac{\beta R}{\beta R + \eta}$$

$$C = \frac{\alpha 1 + \alpha 2}{\beta R + \eta}$$

The Key Threshold Condition

The entire long-term behavior of the system depends on comparing A and B:

Condition	Outcome
A < B	Sensitive bacteria die out. Only resistant bacteria and immune cells persist at equilibrium E2. Antibiotics have failed to prevent resistance dominance.
A > B	Both sensitive and resistant bacteria coexist with immune cells at equilibrium E3. Neither type is eliminated.

A < B → E2 is stable → resistant bacteria dominate

A > B → E3 is stable → coexistence of both types

Numerical Verification for Both Cases

Case I — A < B

Parameters: $\beta S=0.8$, $\beta R=0.4$, $\eta=0.3$, $\alpha 1=0.02$, $d1=0.15$, $\alpha 2=0.06$, $d2=0.35$

$A = (0.8 - (0.02+0.15) - (0.06+0.35)) / (0.8 + 0.3)$

$= (0.8 - 0.17 - 0.41) / 1.1$

$= 0.22 / 1.1$

$= 0.2000$

$B = 0.4 / (0.4 + 0.3)$

$= 0.4 / 0.7$

$= 0.5714$

$A = 0.2000 < B = 0.5714$ → E2 is stable

Case II — A > B

Parameters: $\beta S=0.8$, $\beta R=0.1$, $\eta=0.05$, $\alpha 1=0.001$, $d1=0.01$, $\alpha 2=0.001$, $d2=0.01$

$A = (0.8 - 0.011 - 0.011) / (0.8 + 0.05)$

$= 0.778 / 0.85$

$= 0.9153$

$B = 0.1 / (0.1 + 0.05)$

$= 0.1 / 0.15$

$= 0.6667$

$A = 0.9153 > B = 0.6667$ → E3 is stable

Case	A	B	Condition	Stable Equilibrium
Case I	0.2000	0.5714	A < B	E2 — resistant bacteria dominate
Case II	0.9153	0.6667	A > B	E3 — coexistence

2.3 Mathematical Analysis

2.3.1 Equilibrium Points

Finding Equilibrium Points

Equilibrium points are found by setting all derivatives equal to zero and solving the resulting algebraic system.

$\frac{ds}{dt} = 0$

$\frac{dr}{dt} = 0$

$\frac{db}{dt} = 0$

$\frac{dai}{dt} = 0$ for i = 1, 2

Then From $\frac{dai}{dt} = 0$

- ⇨ $\mu i \cdot (1 - ai) = 0$
- ⇨ $ai = 1$ for i = 1, 2

All equilibrium points have a1 = a2 = 1. Substituting ai = 1 into the remaining three equations reduces the system to:

$\beta S \cdot s \cdot (1-s-r) - \eta \cdot s \cdot b - s \cdot \Sigma(\alpha i + di) = 0$ ... (i)

$\beta R \cdot r \cdot (1-s-r) - \eta \cdot r \cdot b + s \cdot \Sigma \alpha i = 0$ ... (ii)

$k \cdot b \cdot (1 - b/(s+r)) = 0$ ... (iii)

From equation (iii) either:

b = 0 (no immune response) or b = s + r (immune cells at maximum capacity)

Case 1 — When b = 0

Substituting b = 0 into (i) and (ii):

$s \cdot [\beta S \cdot (1-s-r) - \Sigma(\alpha i + di)] = 0$

$r \cdot [\beta R \cdot (1-s-r)] = 0$

This gives $s = 0$ and $r = 0$

Subcase $s = 0, r = 0$ → gives E0

Subcase $s = 0, r \neq 0$ → from $\beta R\cdot(1-r) = 0$ → $r = 1$ → gives E1

Subcase $s \neq 0, r \neq 0$ → gives a point where s and r have opposite signs which is biologically meaningless. so, its rejected.

Case 2 — When $b = s + r$

Substituting $b = s+r$ into (i) and (ii):

$s \cdot [\beta S\cdot(1-s-r) - \eta\cdot(s+r) - \Sigma(\alpha i+di)] = 0$ ... (iv)

$r \cdot [\beta R\cdot(1-s-r) - \eta\cdot(s+r)] + s\cdot\Sigma\alpha i = 0$ ... (v)

Subcase $s = 0$ → substituting into (v):

$r \cdot [\beta R\cdot(1-r) - \eta\cdot r] = 0$

$r \cdot [\beta R - (\beta R+\eta)\cdot r] = 0$

$r = \beta R/(\beta R+\eta) = B$ → gives E2

Subcase $s \neq 0$ → from (iv) the bracket equals zero:

$\beta S\cdot(1-s-r) - \eta\cdot(s+r) - \Sigma(\alpha i+di) = 0$

$(\beta S+\eta)\cdot(1-s-r) = \beta S - \Sigma(\alpha i+di)$

$(s+r) = [\beta S - \Sigma(\alpha i+di)]/(\beta S+\eta) = A = \bar{b}$

Substituting back gives $\bar{s}$ and $\bar{r}$ → gives E3

Equilibrium	**Values**	**Name**
E0	(0, 0, 0, 1, 1)	Infection free
E1	(0, 1, 0, 1, 1)	Resistant bacteria only
E2	(0, B, B, 1, 1)	Resistant bacteria and immune cells
E3	($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1)	Coexistence of all populations

Existence Conditions

Equilibrium	Exists When	Biological Meaning
E0	Always	System can always be infection free in theory
E1	Always	Full resistant bacteria dominance always possible
E2	Always	Resistant bacteria and immune coexistence always possible
E3	A > B only	Coexistence only when sensitive bacteria are relatively strong

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Key Observation

E0 and E1 always exist but as shown in Chapter 4.2 they are always unstable — the system never stays at these points. The system always moves toward either E2 or E3 depending on whether A < B or A > B.

2.3.2 Stability Analysis

The Jacobian Matrix

stability is determined by linearizing the system around each equilibrium point. This gives the Jacobian matrix J evaluated at each equilibrium:

$$J = \begin{bmatrix} \frac{\partial \varphi 1}{\partial s} & \frac{\partial \varphi 1}{\partial r} & \frac{\partial \varphi 1}{\partial b} & \frac{\partial \varphi 1}{\partial a1} & \frac{\partial \varphi 1}{\partial a2} \\ \frac{\partial \varphi 2}{\partial s} & \frac{\partial \varphi 2}{\partial r} & \frac{\partial \varphi 2}{\partial b} & \frac{\partial \varphi 2}{\partial a1} & \frac{\partial \varphi 2}{\partial a2} \\ \frac{\partial \varphi 3}{\partial s} & \frac{\partial \varphi 3}{\partial r} & \frac{\partial \varphi 3}{\partial b} & \frac{\partial \varphi 3}{\partial a1} & \frac{\partial \varphi 3}{\partial a2} \\ \frac{\partial \varphi 4}{\partial s} & \frac{\partial \varphi 4}{\partial r} & \frac{\partial \varphi 4}{\partial b} & \frac{\partial \varphi 4}{\partial a1} & \frac{\partial \varphi 4}{\partial a2} \\ \frac{\partial \varphi 5}{\partial s} & \frac{\partial \varphi 5}{\partial r} & \frac{\partial \varphi 5}{\partial b} & \frac{\partial \varphi 5}{\partial a1} & \frac{\partial \varphi 5}{\partial a2} \end{bmatrix}$$

Where ϕ1...ϕ5 are the right hand sides of the five ODEs. Evaluated explicitly:

J =

$$\begin{bmatrix} \beta S - 2\beta Ss - \beta Sr - \eta b - \Sigma(\alpha i + di)ai & -\beta Ss & -\eta s & -s(\alpha 1 + d1) & -s(\alpha 2 + d2) \\ -\beta Rr + \Sigma\alpha iai & \beta R - \beta Rs - 2\beta Rr - \eta b & -\eta r & s\alpha 1 & s\alpha 2 \\ kb^2/(s+r)^2 & kb^2/(s+r)^2 \quad k - 2kb/(s+r) & k - 2kb/(s+r) & 0 & 0 \\ 0 & 0 & -\mu 1 & -\mu 1 & 0 \\ 0 & 0 & 0 & 0 & -\mu 2 \end{bmatrix}$$

The eigenvalues λ of J at each equilibrium determine stability:

All eigenvalues have negative real parts → equilibrium is Stable

Any eigenvalue has positive real part → equilibrium is unstable

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Routh-Hurwitz Criteria

For a 5×5 Jacobian, finding eigenvalues directly requires solving a degree-5 polynomial — which has no general closed form solution. The Routh-Hurwitz criteria allow us to determine whether ALL roots have negative real parts without solving for the roots explicitly.

The Criteria

Given characteristic polynomial:

$P(\lambda) = \lambda^n + a1\lambda^{n-1} + a2\lambda^{n-2} + ... + an = 0$

All roots have negative real parts if and only if all Hurwitz determinants are positive.

For n = 2:

Conditions: $a1 > 0$ and $a2 > 0$

For n = 3:

Conditions: $a1 > 0$ and $a3 > 0$ and $a1 \cdot a2 > a3$

Stability of E0

$\lambda 1 = \beta S - \Sigma(\alpha i + di)$ (can be positive)

$\lambda 2 = \beta R > 0$ (always positive)

Since $\lambda 2 = \beta R$ is always positive by definition of parameters, E0 is always unstable. The infection free state cannot be maintained.

Stability of E1

$\lambda 1 = -\Sigma(\alpha i+di) < 0$

$\lambda 2 = -\beta R < 0$

$\lambda 3 = k > 0$ ← always positive

Since $\lambda 3 = k$ is always positive, E1 is always unstable. Pure resistant bacteria without any immune response cannot be maintained.

Stability of E2

Substituting E2 = (0, B, B, 1, 1) into J

The eigenvalues from the last two rows are immediately:

$\lambda 4 = -\mu 1 < 0$

$\lambda 5 = -\mu 2 < 0$

These are always stable. The remaining three eigenvalues come from the block matrix:

$$JB(E2) = \begin{bmatrix} \beta S(1-B) - \eta \cdot B - \Sigma(\alpha i + di) & 0 & 0 \\ -\beta R \cdot B + \Sigma \alpha i & -\beta R \cdot B - \eta \cdot B & 0 \\ k & k & -k \end{bmatrix}$$

The characteristic equation factors as:

$[\beta S(1-B) - \eta B - \Sigma(\alpha i+di) - \lambda] \cdot [\lambda^2 + \lambda(k + \beta R \cdot B) + k \cdot \beta R \cdot B] = 0$

This gives:

$\lambda 2,1 = \beta S(1-B) - \eta B - \Sigma(\alpha i+di)$

$= (\beta S + \eta) \cdot (A - B)$

And $\lambda 2,2$ and $\lambda 2,3$ from:

$\lambda^2 + (k + \beta R \cdot B). \lambda + k \cdot \beta R \cdot B = 0$

Applying Routh-Hurwitz with n=2:

$a1 = k + \beta R \cdot B > 0$ (all parameters positive)

$a2 = k \cdot \beta R \cdot B > 0$ (all parameters positive)

So $\lambda 2,2$ and $\lambda 2,3$ always have negative real parts.

The only condition remaining is the sign of $\lambda 2,1$:

$\lambda 2,1 = (\beta S + \eta)(A - B)$

$A < B \rightarrow A - B < 0 \rightarrow \lambda 2,1 < 0 \rightarrow$ E2 is stable

$A > B \rightarrow A - B > 0 \rightarrow \lambda 2,1 > 0 \rightarrow$ E2 is unstable

Therefore, E2 is stable if and only if A < B

Stability of E3

Substituting E3 = ($\bar{s}$, $\bar{r}$, $\bar{b}$, 1, 1) into J

Again the last two eigenvalues are:

$\lambda 4 = -\mu 1 < 0$

$\lambda 5 = -\mu 2 < 0$

The remaining three eigenvalues come from the block matrix:

$$JB(E3) = \begin{bmatrix} -\beta S \cdot \bar{s} & -\beta S \cdot \bar{s} & -\eta \cdot \bar{s} \\ -\beta R \cdot \bar{r} + \Sigma \alpha i & \beta R(1 - \bar{b}) - \eta \cdot \bar{b} - \beta R \cdot \bar{r} & -\eta \cdot \bar{r} \\ k & k & -k \end{bmatrix}$$

The characteristic equation is:

$\lambda^3 + a1 \cdot \lambda^2 + a2 \cdot \lambda + a3 = 0$

Where:

$a1 = \bar{s} \cdot \bar{r} \cdot \Sigma \alpha i + \beta R \cdot \bar{r} + \beta S \cdot \bar{s} + k$

$a2 = k(\bar{s} \cdot \bar{r} \cdot \Sigma \alpha i + \beta R \cdot \bar{r} + \beta S \cdot \bar{s}) + \bar{b} \cdot \eta \cdot k + \beta S \cdot \bar{b} \cdot \bar{s} \cdot \bar{r} \cdot \Sigma \alpha i$

$a3 = k \cdot \eta \cdot \bar{b} \cdot \bar{s} \cdot \bar{r} \cdot \Sigma \alpha i + k \cdot \beta S \cdot \bar{b} \cdot \bar{s} \cdot \bar{r} \cdot \Sigma \alpha i$

Applying Routh-Hurwitz with n=3:

$a1 > 0$ (all terms positive since $\bar{s}$, $\bar{r}$, $\bar{b} > 0$ when $A > B$)

$a3 > 0$ (all terms positive)

$a1 \cdot a2 > a3$

All three Routh-Hurwitz conditions are satisfied whenever E3 exists, that is whenever $A > B$.

Therefore, E3 is stable if and only if $A > B$.

Summary Table

Equilibrium	Eigenvalue Signs	Stability Condition
E0	At least one positive	Always unstable
E1	At least one positive	Always unstable
E2	All negative when $A < B$	$A < B$
E3	All negative when $A > B$	$A > B$

Biological Interpretation of Each Equilibrium

Equilibrium	Biological Meaning
E0	Infection free state. Theoretically possible but always unstable — any small bacterial introduction will grow and push the system away from this point.
E1	Resistant bacteria completely dominate with no immune response. Always unstable — any immune activation will disturb this state.
E2	Sensitive bacteria are eliminated by antibiotics. Only resistant bacteria remain, held in balance by the immune system. This is the worst clinical outcome — resistance has won.

Equilibrium	Biological Meaning
E3	Both sensitive and resistant bacteria coexist with active immune response. Neither type is fully eliminated. Antibiotics alone are insufficient to clear the infection.

2.4 Results and Discussion

2.4.1 Parameter Values Used

All parameter values used in the numerical simulations are listed below.

Biological Parameters

Symbol	Description	Value	Unit	Source
βS	Growth rate of sensitive bacteria	0.8	day^{-1}	Mondragón et al. 2014
βR	Growth rate of resistant bacteria	0.4	day^{-1}	Mondragón et al. 2014
c	Fitness cost of resistance	0.5	dimensionless	Hypothesis
η	Immune killing rate	0.3	day^{-1}	Pugliese and Gandolfi 2008
k	Immune cell proliferation rate	0.6	day^{-1}	Pugliese and Gandolfi 2008
ω	Ratio of immune cell capacity to bacterial capacity	1	dimensionless	Hypothesis
T	Carrying capacity of bacteria	10^9	cells	Alavez et al. 2006

Antibiotic Parameters — Case I (A < B)

Symbol	Description	Value	Unit
$\alpha 1$	Mutation rate due to INH	0.02	day^{-1}
d1	Killing rate of sensitive bacteria by INH	0.15	day^{-1}
$\mu 1$	Clearance rate of INH	0.06	day^{-1}
$\alpha 2$	Mutation rate due to PZA	0.06	day^{-1}
d2	Killing rate of sensitive bacteria by PZA	0.35	day^{-1}
$\mu 2$	Clearance rate of PZA	0.03	day^{-1}

Antibiotic Parameters — Case II (A > B)

Symbol	Description	Value	Unit
βR	Growth rate of resistant bacteria	0.1	day^{-1}
η	Immune killing rate	0.05	day^{-1}
$\alpha 1$	Mutation rate due to INH	0.001	day^{-1}
d1	Killing rate of sensitive bacteria by INH	0.01	day^{-1}
$\alpha 2$	Mutation rate due to PZA	0.001	day^{-1}
d2	Killing rate of sensitive bacteria by PZA	0.01	day^{-1}
$\mu 1$	Clearance rate of INH	0.06	day^{-1}
$\mu 2$	Clearance rate of PZA	0.03	day^{-1}

Initial Conditions

Variable	Description	Value	Justification
s(0)	Initial sensitive bacteria	0.80	Heavy bacterial load at start of treatment
r(0)	Initial resistant bacteria	0.05	Small resistant population already present
b(0)	Initial immune cells	0.05	Weak immune response at start
a1(0)	Initial INH concentration	0.05	Antibiotic just administered
a2(0)	Initial PZA concentration	0.02	Antibiotic just administered

---

2.4.2 Threshold Values A, B and C

Computed Values for Both Cases

Parameter	Formula	Case I	Case II
A	$(\beta S - \Sigma(\alpha i+di)) / (\beta S+\eta)$	0.2000	0.9259
B	$\beta R / (\beta R+\eta)$	0.5714	0.6667
C	$\Sigma\alpha i / (\beta R+\eta)$	0.1143	0.0267
Condition	—	A < B	A > B
Stable Equilibrium	—	E2	E3

Interpretation

In Case I where A < B, the antibiotics and immune system together are strong enough to eliminate sensitive bacteria completely. However resistant bacteria survive because they are unaffected by antibiotics and can only be controlled by the immune system. The system converges to E2 where only resistant bacteria and immune cells coexist at equilibrium. This represents the clinically unfavorable outcome where antibiotic resistance has effectively won.

In Case II where A > B, the antibiotics are too weak relative to the growth advantage of sensitive bacteria, or the resistant bacteria have a very high fitness cost making them slow to grow. Neither bacterial type is fully eliminated and the system converges to E3 where sensitive bacteria, resistant bacteria, and immune cells all coexist at a stable positive equilibrium.

2.4.3 Case I Results — A < B

All simulations in this section use the Case I parameters where A = 0.2000 and B = 0.5714, giving the condition A < B. The system is therefore expected to converge to equilibrium E2 = (0, 0.5714, 0.5714, 1, 1).

The following plot shows the time evolution of all five state variables under Case I parameters where A = 0.2000 < B = 0.5714.

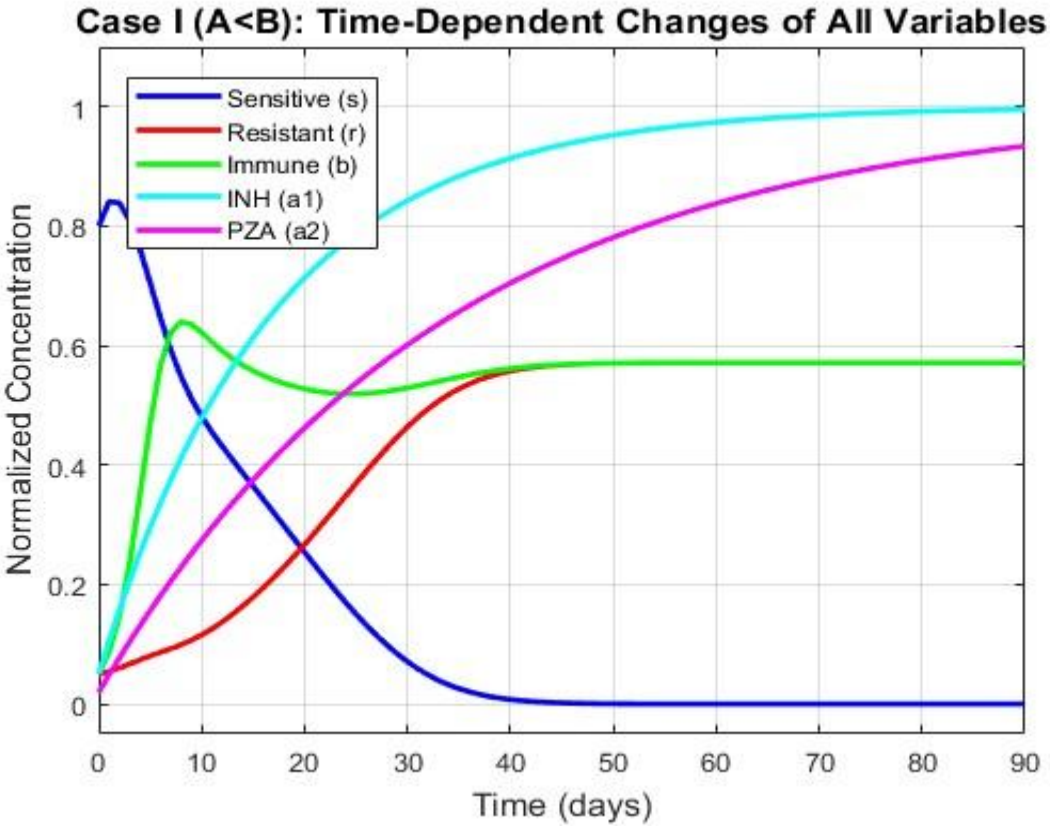

Fig: 2.1 All 5 Variables Over Time (A<B)

Fig: 2.1 — All 5 Variables Over Time Sensitive bacteria steadily decline and reach zero by around day 60, wiped out by antibiotics and the immune system. Resistant bacteria survive and stabilize at a low level. Immune cells rise and settle at the same level as resistant bacteria. Both antibiotics quickly reach their maximum concentration and stay there. Overall, the system settles at equilibrium E_2 exactly as the math predicted.

The immune cell spike visible in the first 5 days reflects rapid proliferation in response to the high initial bacterial load.

The following plot isolates the relationship between total bacterial load and immune cell population over time.

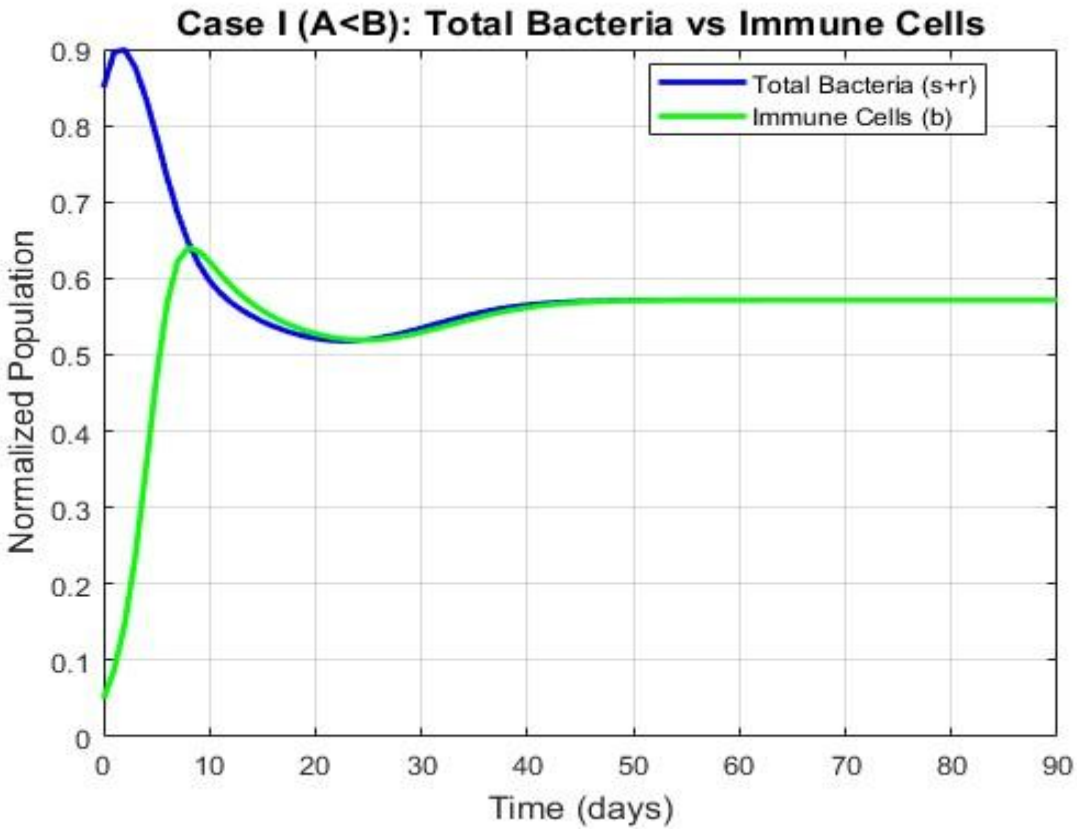

Fig: 2.2 Total Bacteria vs Immune Cells

Fig: 2.2 — Total Bacteria vs Immune Cells Shows that the immune system tracks and matches the bacterial load over time. By day 90 both curves meet at the same value, confirming that the immune system brings the infection under control — not eliminated, but held in a stable balance.

The following plot shows the pharmacokinetic behavior of both antibiotics over the 200-day treatment period.

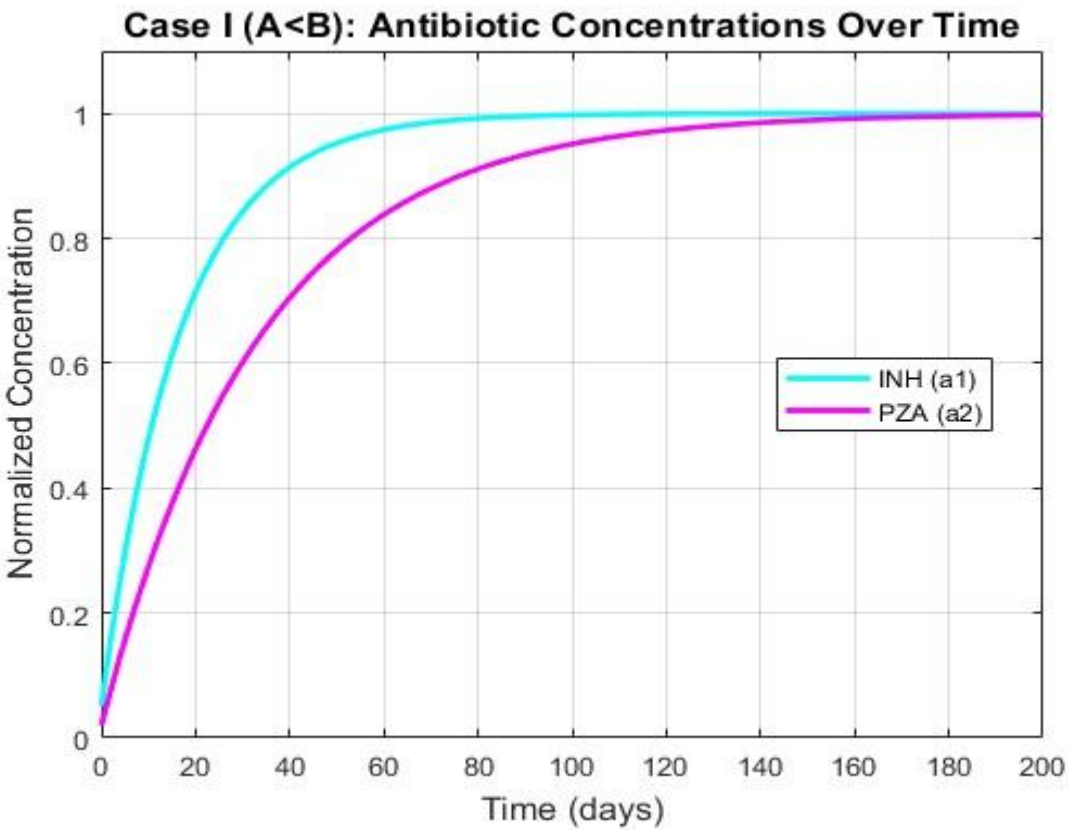

Fig: 2.3 Antibiotic Concentrations

Fig: 2.3 — Antibiotic Concentrations Both drugs rise quickly and plateau within the first 30 days. INH stabilizes faster than PZA due to its higher clearance rate. After that, concentrations stay constant for the rest of the treatment.

The following phase portrait shows the trajectory of sensitive bacteria s against resistant bacteria r over the 90-day period under Case I conditions.

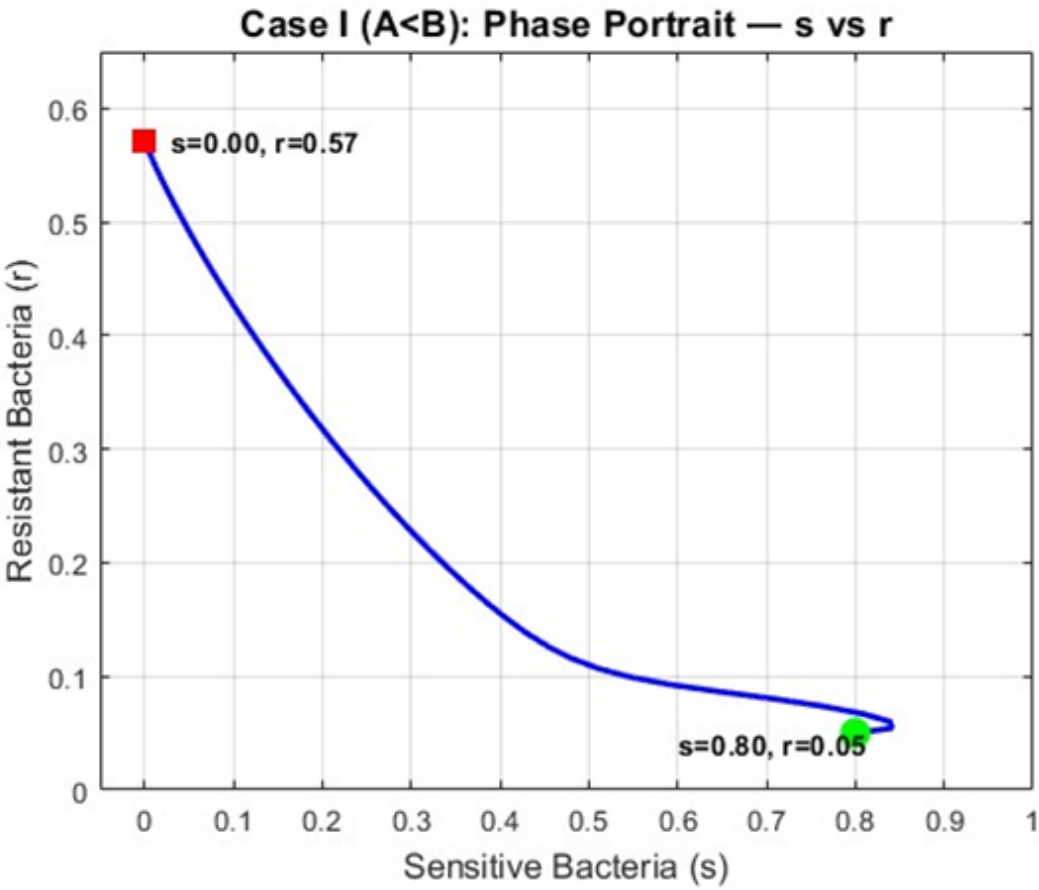

Fig: 2.4 Phase Portrait (s vs r)

The trajectory moves from the initial condition at s=0.80, r=0.05 toward the E2 equilibrium at s=0, r=0.5714. By day 90 sensitive bacteria have reached zero and resistant bacteria have stabilized at 0.5714, confirming complete convergence to E2. Comparing this with the Case II phase portrait in Section 6.4 clearly shows the difference — in Case I the trajectory terminates on the r-axis where s=0, while in Case II the trajectory terminates at a positive s value confirming coexistence.

Numerical Results — Case I (A < B)

The following table records exact numerical values at 10-day intervals confirming convergence to E2 = (0, 0.5714, 0.5714, 1, 1).

Time (days)	Sensitive (s)	Resistant (r)	Total (s+r)	Immune Cells (b)	INH (a1)	PZA (a2)
0	0.8000	0.0500	0.8500	0.0500	0.0500	0.0200
10	0.4812	0.1157	0.5969	0.6219	0.4786	0.2740
20	0.2542	0.2671	0.5213	0.5275	0.7139	0.4622
30	0.0708	0.4641	0.5349	0.5294	0.8430	0.6016
40	0.0075	0.5577	0.5651	0.5617	0.9138	0.7048
50	0.0004	0.5706	0.5710	0.5707	0.9527	0.7813
60	0.0000	0.5714	0.5714	0.5714	0.9740	0.8380
70	0.0000	0.5714	0.5714	0.5714	0.9858	0.8800
80	0.0000	0.5714	0.5714	0.5714	0.9922	0.9111
90	0.0000	0.5714	0.5714	0.5714	0.9957	0.9341

Table: 2.1 Numerical Results — Case I (A < B)

Sensitive bacteria fall to zero by day 60 and remain there. Resistant bacteria and immune cells both stabilize at 0.5714 by day 90, numerically confirming the theoretical equilibrium E2.

2.4.4 Case II Results — A > B

All simulations in this section use the Case II parameters where A = 0.3345 and B = 0.2500, giving the condition A > B. The system is therefore expected to converge to equilibrium E_3 = (0.3337, 0.0008, 0.3345, 1, 1).

The following plot shows the time evolution of all five state variables under Case II parameters where A = 0.3364 > B = 0.2500.

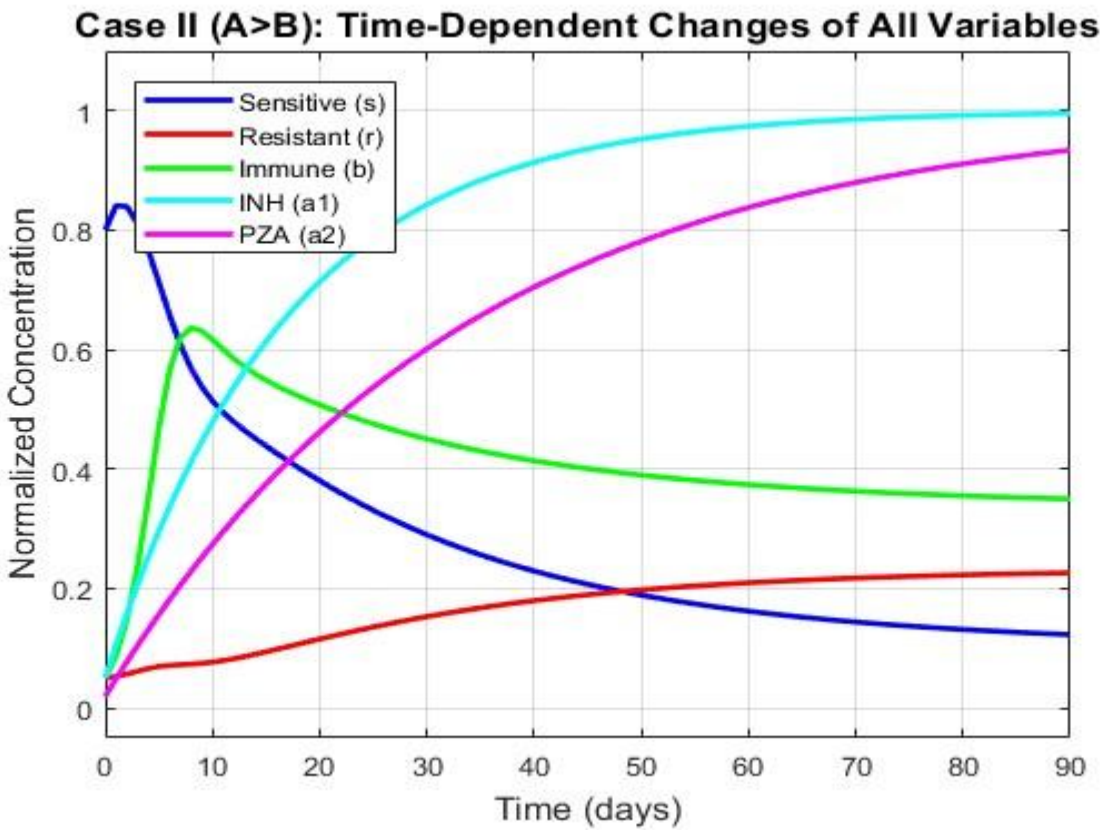

Fig: 2.5 All 5 Variables Over Time (A>B)

Fig: 2.5 — All 5 Variables Over Time Unlike Case I, sensitive bacteria do not go to zero — they decline but stabilize at a positive value. Resistant bacteria also persist. Both coexist alongside immune cells at a stable equilibrium, confirming convergence to E_3 as predicted by the model.

The following phase portrait shows the trajectory of sensitive bacteria s against resistant bacteria r over the 90-day period.

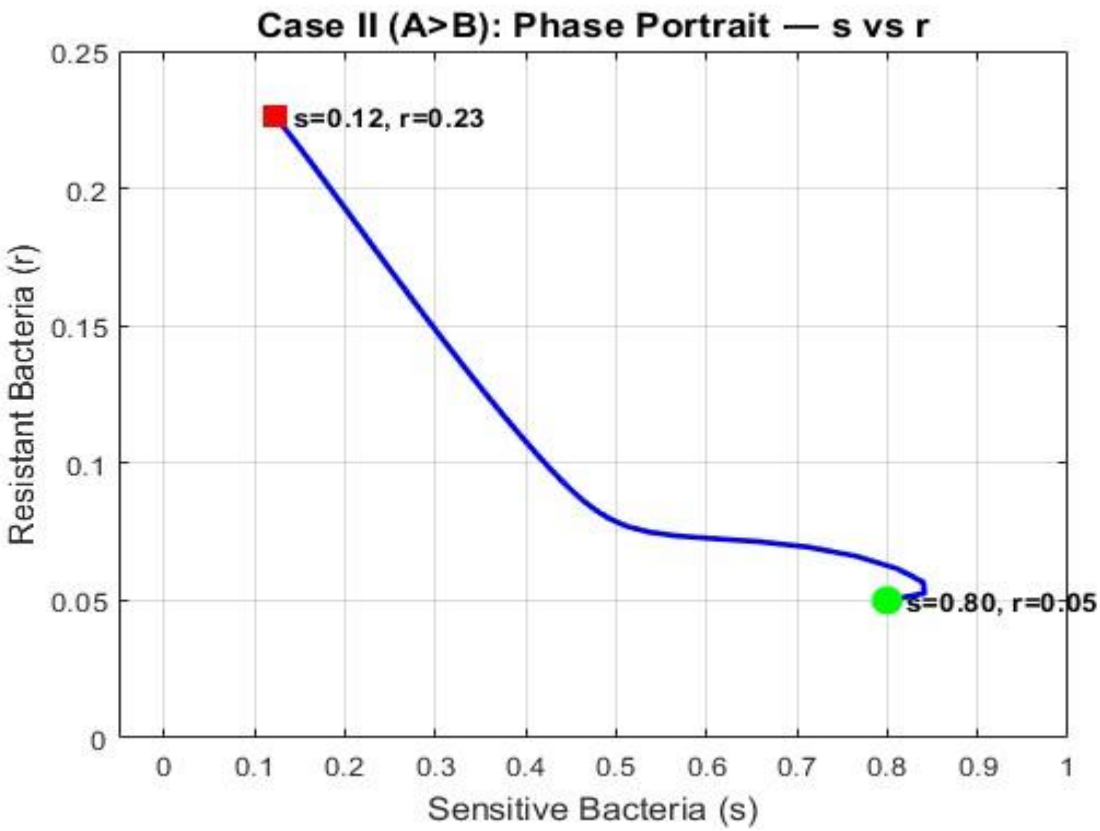

Fig: 2.6 Phase Portrait (s vs r)

Fig: 2.6 — Phase Portrait (s vs r) Shows the trajectory of the system from start (green) to end (red). The path curves toward E_3 where both s and r are positive, visually confirming that neither bacterial strain is eliminated under these conditions.

The following plot shows all three biological populations together to demonstrate coexistence under Case II conditions.

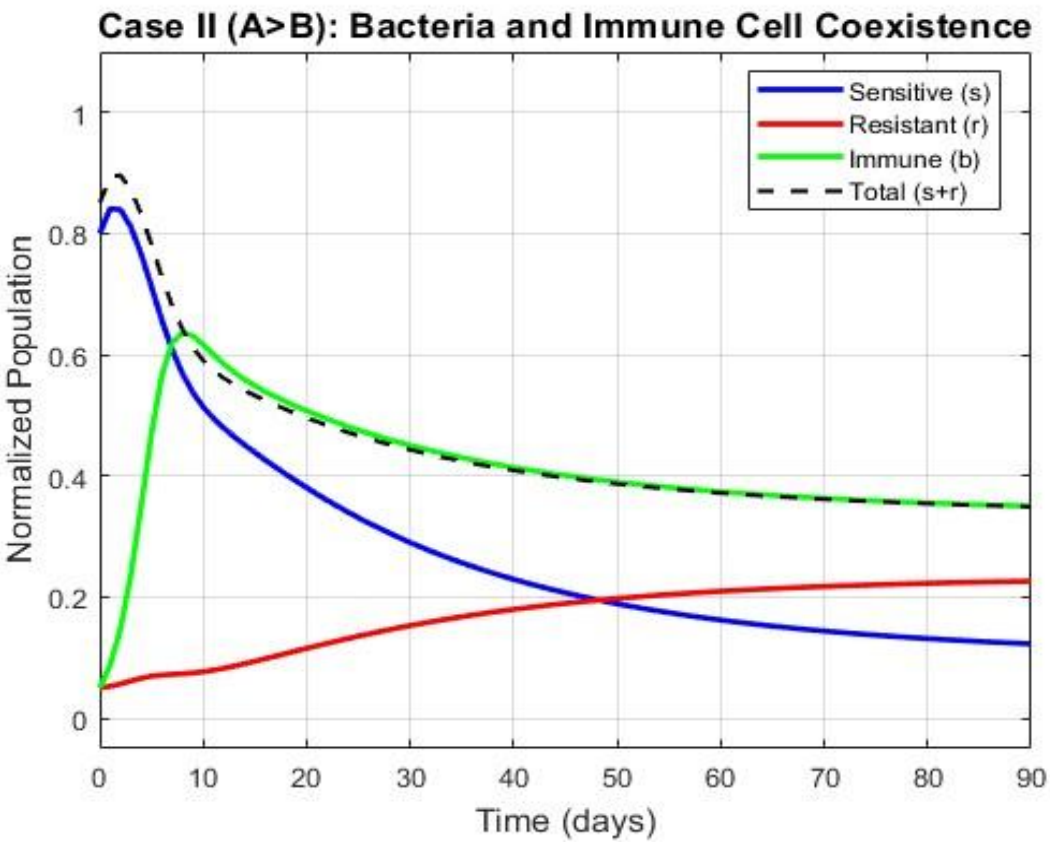

Fig: 2.7 Bacteria and Immune Coexistence

Fig: 2.7 — Bacteria and Immune Coexistence All three populations — sensitive, resistant, and immune cells — stabilize at positive values by day 90. The immune system cannot fully clear the infection; instead, a three-way balance is reached. This reflects the biological scenario where antibiotics and immunity together are insufficient to eliminate the bacteria.

Unlike Case I where immune cells converged to the same values resistant bacteria, in Case II immune cells track the total bacterial load s+r which remains higher due to the persistence of sensitive bacteria alongside resistant ones.

Numerical Results — Case II (A > B)

The following table records exact numerical values at 20-day intervals showing convergence toward E3.

Time (days)	Sensitive (s)	Resistant (r)	Total (s+r)	Immune Cells (b)	INH (a1)	PZA (a2)
0	0.8000	0.0500	0.8500	0.0500	0.0500	0.0200
20	0.3805	0.1155	0.4960	0.5075	0.7139	0.4622
40	0.2293	0.1798	0.4091	0.4139	0.9138	0.7048
60	0.1622	0.2096	0.3718	0.3740	0.9740	0.8380
80	0.1315	0.2228	0.3543	0.3554	0.9922	0.9111
100	0.1169	0.2288	0.3457	0.3462	0.9976	0.9512
120	0.1095	0.2318	0.3413	0.3416	0.9993	0.9732
140	0.1058	0.2333	0.3390	0.3392	0.9998	0.9853
160	0.1038	0.2340	0.3378	0.3379	0.9999	0.9919
180	0.1027	0.2344	0.3371	0.3372	1.0000	0.9956
200	0.1021	0.2347	0.3368	0.3368	1.0000	0.9976
220	0.1018	0.2348	0.3366	0.3366	1.0000	0.9987
240	0.1016	0.2348	0.3365	0.3365	1.0000	0.9993

Table: 2.2 Numerical Results — Case II (A > B)

The table records numerical values at 20-day intervals up to day 240. By day 220 the system has effectively converged with changes between consecutive time points less than 0.0002. All three populations s, r and b remain positive throughout — s stabilizes at 0.1016, r at 0.2348, and b at 0.3365 — numerically confirming the theoretical equilibrium Unlike Case I where sensitive bacteria reached zero, in Case II all populations coexist at positive values as predicted by the A > B threshold condition.

2.4.5 Effect of Immune System Strength

To examine how immune system strength affects the outcome, the immune killing rate η was varied across four values (0.1, 0.3, 0.6, 1.0) while all other parameters were held constant at Case I values.

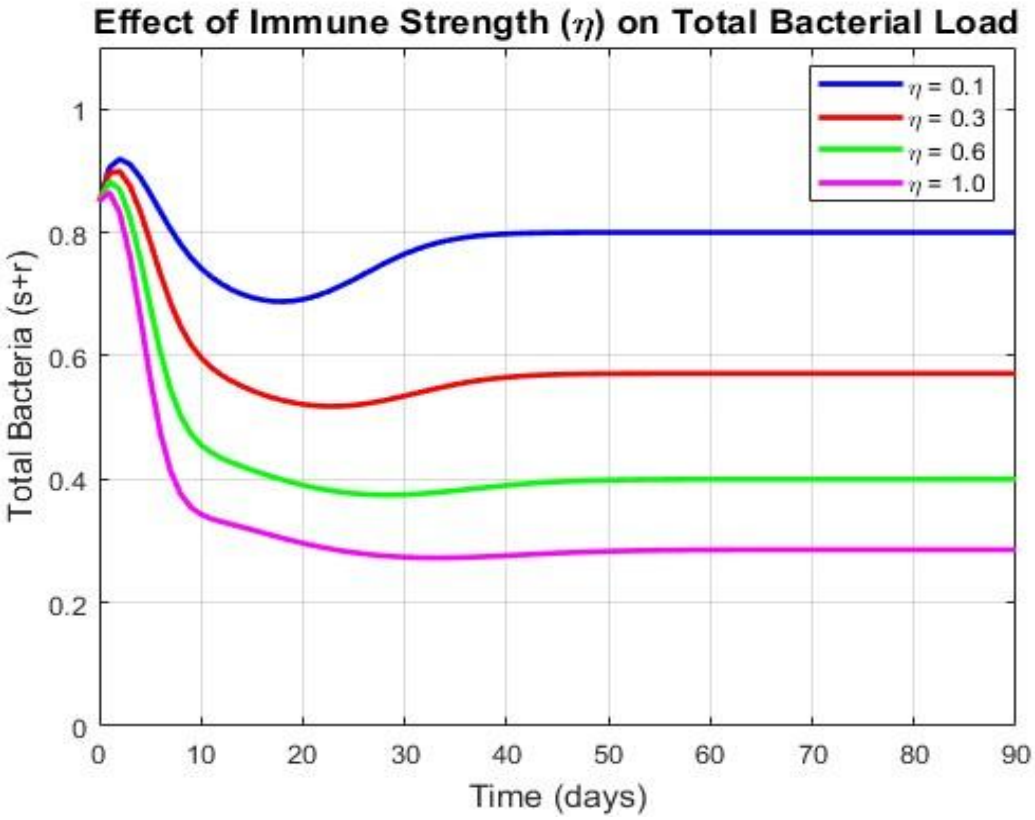

Fig: 2.8 Effect of Immune Strength on Total Bacterial load

The plot shows total bacterial load over 90 days for four values of η (0.1, 0.3, 0.6, 1.0), where η represents the rate at which immune cells kill bacteria.

When η is low (0.1), the immune system is weak and bacterial load remains high throughout the treatment. As η increases, bacteria are cleared faster and settle at a lower equilibrium. At η = 1.0, the bacterial burden is reduced most aggressively. This confirms what the model predicts — a stronger immune response directly lowers the equilibrium bacterial load. In clinical terms, immunocompromised patients (low η) will carry a higher bacterial burden even under the same antibiotic treatment, making recovery harder.

2.4.6 Effect of Fitness Cost

The fitness cost parameter c was varied across four values (0.1, 0.3, 0.5, 0.9) to show its effect on the resistant bacterial population. All other parameters were held at Case I values.

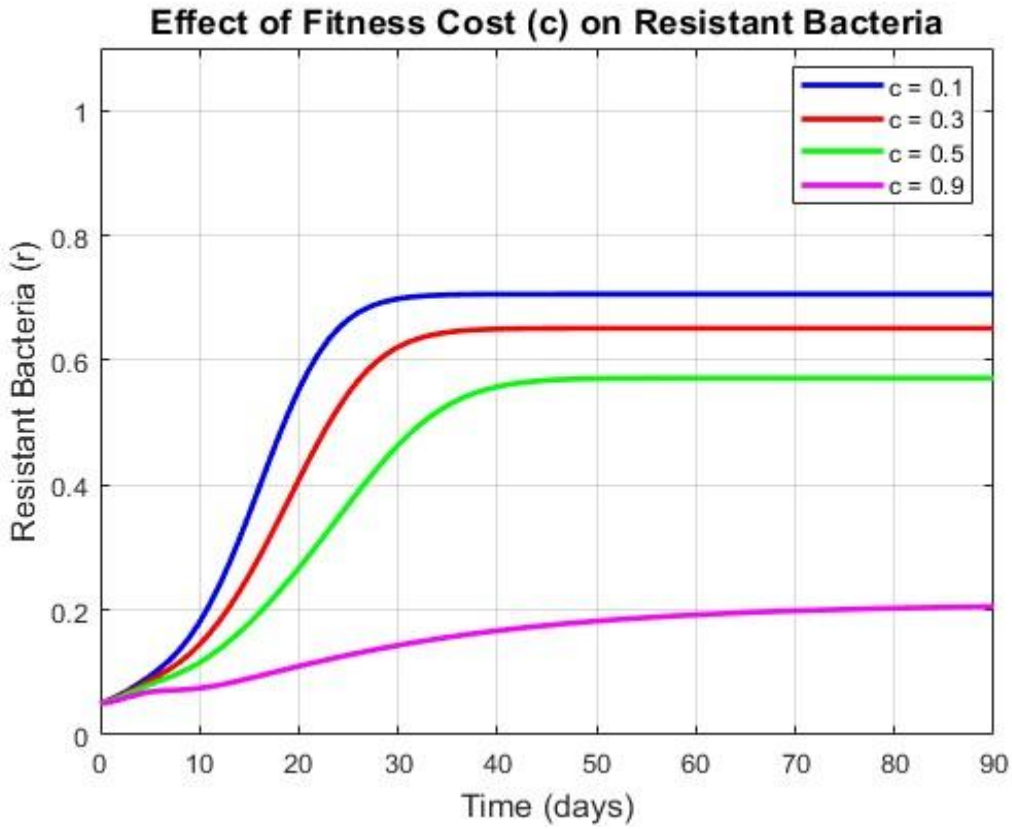

Fig: 2.9 Effect of Fitness Cost in Resistance Bacteria

The plot shows how the resistant bacterial population evolves over 90 days for four values of fitness cost c (0.1, 0.3, 0.5, 0.9), where c directly reduces the growth rate of resistant bacteria as $\beta_R = (1-c)\,\beta_S$.

When c is small (0.1), resistant bacteria carry almost no fitness penalty and grow rapidly to a high equilibrium level. As c increases, resistant bacteria grow more slowly and stabilize at progressively lower levels. At c = 0.9, the fitness cost is severe and resistant bacteria remain at a very low level throughout. This shows that the biological cost of resistance is a natural limiting factor — the higher the cost, the less dominant resistant strains become. In clinical terms, this suggests that reducing antibiotic pressure can allow high-cost resistant strains to die out naturally over time.

2.4.7 Early Antibiotic Termination

To model the clinical scenario of incomplete treatment, antibiotic supply was set to zero after day 30, day 60, and kept active for the full 90 day course respectively.

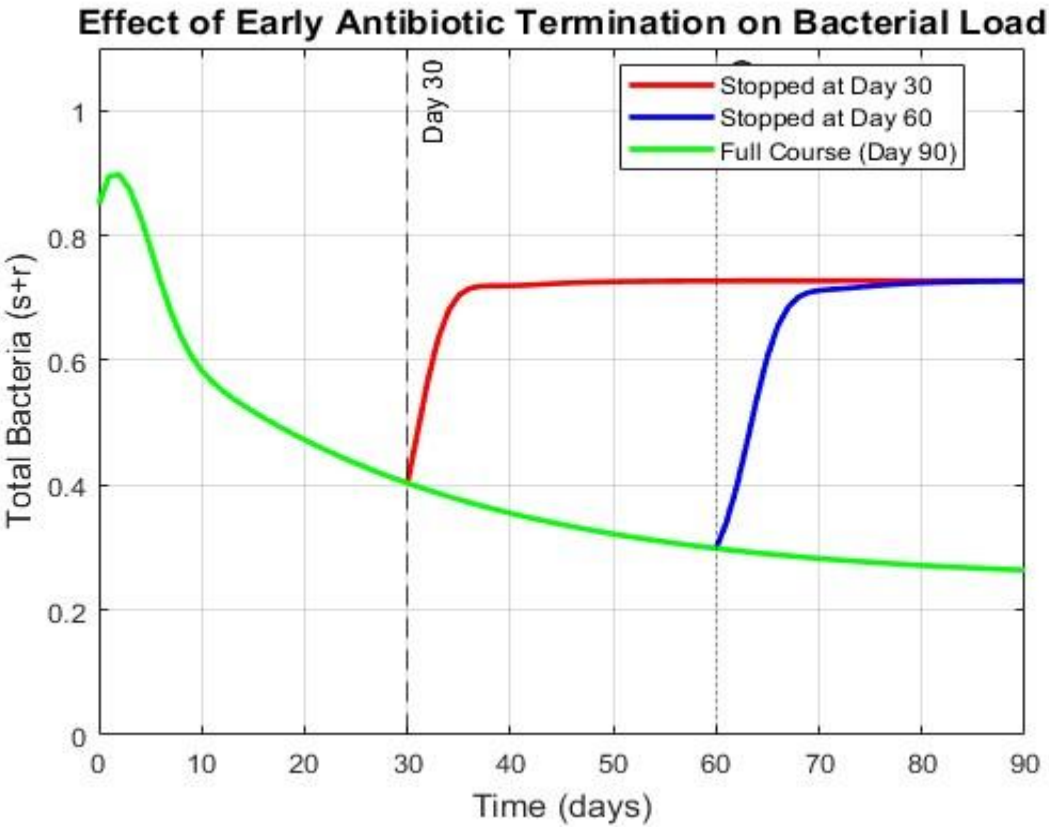

Fig: 2.10 Effect of Antibiotic Termination on Total Bacterial load

The plot compares total bacterial load when antibiotics are stopped at day 30, day 60, or completed for the full 90-day course.

When treatment is stopped at day 30, bacteria rebound sharply after the antibiotics are removed. The immune system alone is insufficient to control the infection and bacterial load climbs back toward high levels. Stopping at day 60 shows a partial rebound — bacteria had been suppressed further before treatment ended, so the rise is slower and settles at a higher level than the full course but lower than the day 30 case. Completing the full 90-day course produces the lowest final bacterial burden, with the system held at the E_2 equilibrium throughout. This result has a direct clinical message — incomplete antibiotic courses not only fail to clear the infection but actively create

conditions for bacterial resurgence, and in practice, for resistance amplification.

2.4.8 Single vs Combination Therapy

The following plot shows sensitive bacteria s only, which more clearly demonstrates the difference in clearance speed between the three treatment strategies.

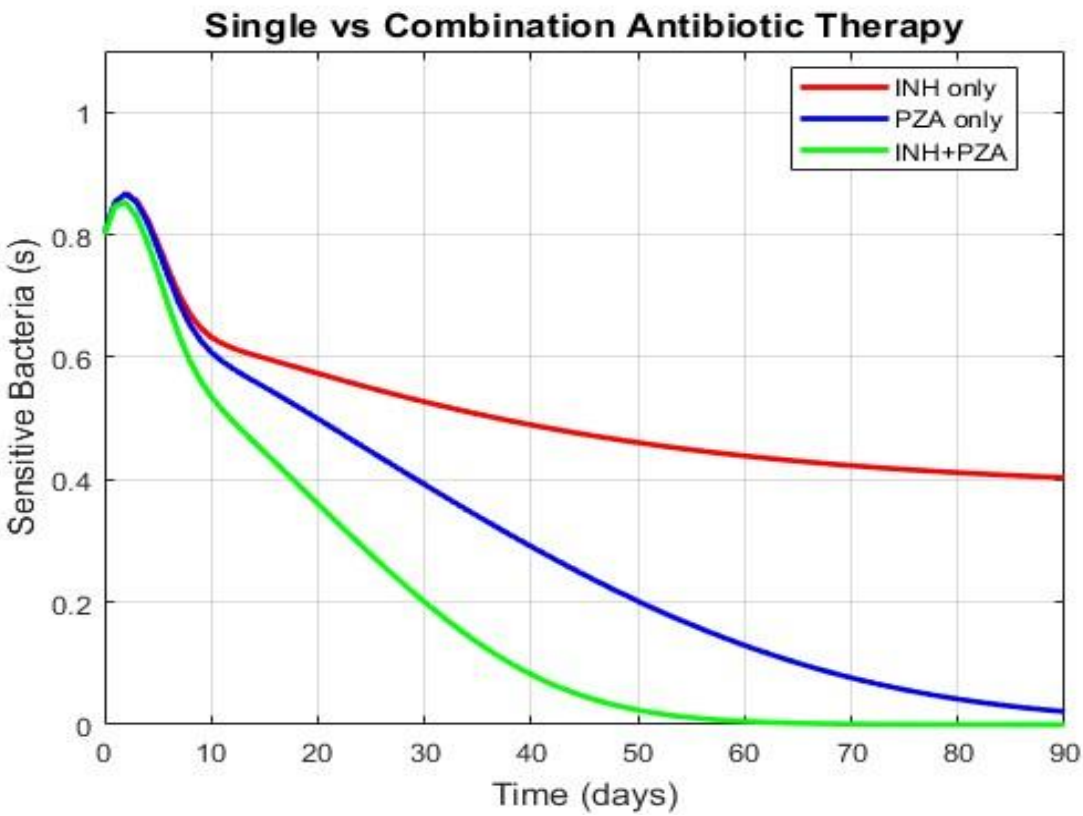

Fig: 2.11 Effect of Single vs Combination Therapy on sensitive bacteria

INH alone fails to eliminate sensitive bacteria by day 90, with s stabilizing at approximately 0.40 — meaning nearly half the sensitive bacterial population survives despite 90 days of treatment. PZA alone eliminates sensitive bacteria by approximately day 85. Combination therapy achieves the fastest clearance, driving sensitive bacteria to zero by approximately day 60 — 25 days faster than PZA alone and far more effective than INH alone. This demonstrates the clear clinical advantage of combination therapy: faster clearance of sensitive bacteria reduces the window during which mutation to resistance can occur, lowering the long-term risk of resistance dominance.

The following plot compares total bacterial load under three treatment strategies using identical initial conditions.

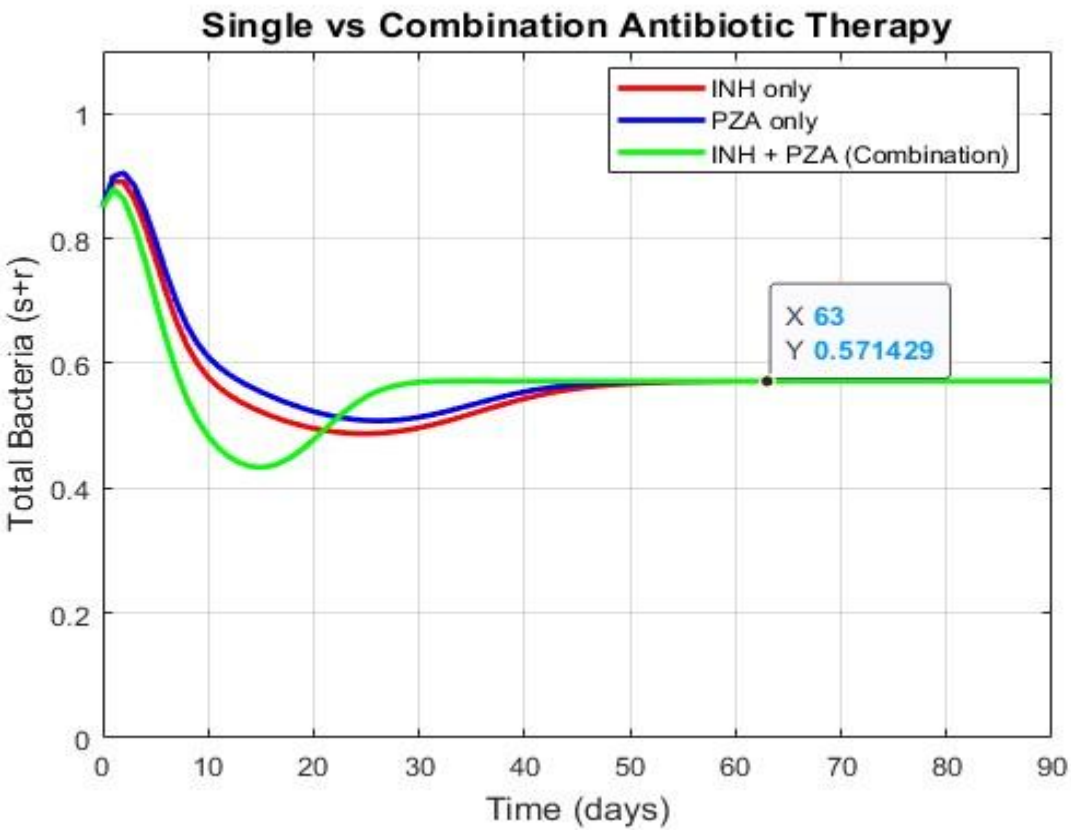

Fig: 2.12 Effect of Single vs Combination Therapy on Total bacteria

The plot compares total bacterial load over 90 days under three treatment strategies: INH alone, PZA alone, and both antibiotics together.

Both single-drug treatments reduce bacterial load but stabilize at a higher equilibrium compared to combination therapy. INH alone performs better than PZA alone due to its stronger killing rate ($d1 > d2$), but neither single drug suppresses bacteria as effectively as the combination.

The data cursor confirms all three therapies converge to the same equilibrium value of 0.5714 at day 63, consistent with the theoretical value $B = \beta R/(\beta R+\eta) = 0.4/0.7 = 0.5714$.

2.4.9 Summary of Results

In Case I where A = 0.2000 < B = 0.5714, the system converged to E2 as predicted. Sensitive bacteria were eliminated by day 60 and resistant bacteria stabilized at 0.5714, matching the theoretical equilibrium exactly. In Case II where A = 0.3364 > B = 0.2500, the system converged to E3 with all three populations stabilizing at positive values — s = 0.1016, r = 0.2348, b = 0.3365 — matching theoretical E3 to four decimal places by day 220.

The sensitivity analyses produced four consistent findings. First, stronger immune response directly reduced equilibrium bacterial load — patients with weakened immunity carried significantly higher bacterial burdens under identical treatment. Second, higher fitness cost of resistance limited resistant bacterial growth, suggesting that reducing antibiotic pressure can allow costly resistant strains to decline naturally. Third, early termination of antibiotics caused sharp bacterial rebound — stopping at day 30 resulted in bacterial load more than twice that of the full 90-day course. Fourth, combination therapy eliminated sensitive bacteria 25 days faster than PZA alone and far more effectively than INH alone, reducing the window during which resistance mutations can emerge.

Across all simulations the outcome was determined by the single threshold comparison A versus B, confirming the predictive value of the model for understanding antibiotic resistance dynamics within a single host.